MANTHAN DHANRAJ

Full-Stack & AI Developer | Computer Science Engineering Student

+91-8789391172 dhanrajmanthan345@gmail.com

Linkedin· Github· LeetCode

Portfolio: **Proof Of Work** **Live:** **AthletixAI**

Professional Summary

Computer Science Engineering student and hands-on builder who ships real, deployed products. Skilled in full-stack development with Python and Flask, machine learning on real-world datasets, and computer-vision pipelines. Comfortable owning projects end-to-end — from architecture and model training to production deployment.

Skills

Languages: Python, C, SQL

Web & Backend: Flask, FastAPI, Jinja2, REST APIs, HTML5, CSS3, SQLite, PostgreSQL (Supabase), Gunicorn

AI / ML & Vision: scikit-learn, Random Forest, MediaPipe (pose), OpenCV, COLMAP (SfM / 3D reconstruction), faster-whisper (STT), Piper (neural TTS), Ollama (local LLMs)

Data & Analysis: pandas, NumPy, feature engineering, model evaluation

Tools & Deployment: Git, GitHub, VS Code, Render, CUDA/GPU, Tkinter, MS Office

Core CS: Data Structures & Algorithms, OOP, File I/O

Projects

AthletixAI — AI Sports Biomechanics & Talent Assessment Platform | *Flask, MediaPipe, SQLite* **Live**

- Built and deployed a live web app that analyzes athletic technique across 48 sports using AI-powered biomechanics scoring for talent assessment.
- Engineered a computer-vision pipeline with MediaPipe 33-keypoint pose analysis to evaluate movement and generate per-sport performance feedback.
- Added trilingual support (English/Hindi/Hinglish), multi-user persistence, secure Gmail SMTP email, and an owner/admin panel with a premium charcoal-gold UI.
- Tuned production performance (SQLite WAL, Gunicorn) and deployed on Render, owning the full stack. Live at athletixai.onrender.com.

JARVIS OS — AI Voice Assistant for Windows 11 | *Python, Ollama, Piper, faster-whisper* · GitHub **v0.7.0**

- Building a JARVIS-style AI operating-system assistant (77 Python modules) with full voice control in Hindi, English, and Hinglish.
- Implemented offline speech-to-text (faster-whisper + CUDA), Piper neural text-to-speech, wake-word detection, and risk-based command confirmation gating.
- Integrated a local LLM (Ollama, qwen2.5) for Hinglish intent understanding with SQLite-backed memory and context awareness.
- Built Windows automation, browser/coding/gaming modes, safe file management, and a Tkinter GUI dashboard, all with safety-first, env-gated power controls.

SOHelioraX — Solar Flare Nowcasting (ISRO BAH 2026) | *Flask, scikit-learn* · GitHub **Team Lead**

- Led a 3-person team to build an ML-powered web app that forecasts solar flares using real Aditya-L1 satellite data (SoLEXS + HEL1OS).
- Trained a Random Forest model achieving ~97.7% cross-validated accuracy across 739 days of space-weather data.
- Built a live dashboard with real-time alerts and email subscriptions; authored a 23-page technical report and pitch deck.

Drone3D Mapper — Drone Video to 3D Reconstruction (SIH 2026) | *FastAPI, OpenCV, COLMAP, Three.js* **In Progress**

- Building a system that turns a single-pass drone video into an accurate 3D model through a full photogrammetry pipeline.
- Engineered the reconstruction spine: frame extraction with quality filtering → COLMAP Structure-from-Motion → dense stereo → mesh → interactive Three.js 3D viewer.
- Built a FastAPI + OpenCV backend, offloading GPU-heavy dense reconstruction to a dedicated CUDA machine.

DSA Implementation Library | *C* — Implemented core data structures (stack, queue, linked list) and search/sort algorithms with documented time/space complexity.

Education

Sharda University, Greater Noida, India **2025 – 2029**

Bachelor of Technology in Computer Science Engineering — CGPA: 8.28 / 10

Achievements & Certifications

Hackathons: OpenAI Codex Hackathon (NamasteDev) — top 30% of 2,989 participants; ChatGPT Codex Hackathon 2026 (BlockseBlock); ISRO Bharatiya Antariksh Hackathon 2026 (Hack2skill) — Certificate of Participation for successful idea submission.

Certifications: Python Programming • Android App Development • AI Tools & ChatGPT Workshop • Google Analytics Fundamentals • Digital Marketing Fundamentals • Project Management Fundamentals

Activities & Interests

Build and ship side projects, solve DSA problems on online judges, and explore open-source production code.